

Samuele TONDELLI

Computer Science and Engineering Student

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“ SUMMARY

Final-year master’s student specializing in machine learning, systems programming, and computer architecture. Focused on optimizing intelligent systems through efficient, hardware-aware software design. Active interest in cybersecurity with hands-on experience in CTF competitions. Seeking internship and research opportunities.

EDUCATION

|                |                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - Present | Master in Computer Science and Engineering, <b>Politecnico di Milano</b>                                                                                                                                                 |
| 2021 - 2024    | Bachelor in Computer Engineering, <b>University of Modena and Reggio Emilia</b> . Thesis : "Optimization and synthesis on FPGA of the logarithm of softmax function for neuromorphic accelerators". <b>110 cum laude</b> |

SKILLS

|               |                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------|
| Languages     | C, Python, Vitis HLS, Rust, CUDA, Verilog, C++, Assembly, Java, Go, JS, HTML/CSS              |
| Embedded      | Xilinx Zynq FPGAs, STM32, Arduino, Raspberry Pi                                               |
| ML frameworks | Scikit-learn, Hydra, PyTorch                                                                  |
| Dev tools     | VS Code, Vim, Vitis, Quartus, ModelSim, Linux, Git, Make, CMake, GCC, GDB, Valgrind, PwnTools |

EXPERIENCE

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|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Present<br>Sep 2025  | <b>Student Researcher   Genomic ML, NECSTLAB, POLITECNICO DI MILANO, Italy</b> <ul style="list-style-type: none"><li>Conducting survey study on ML approaches for single nucleotide variant classification.</li><li>Benchmarked 4 models (MLP, XGBoost, Logistic Reg, Random Forest) across 144 genomic datasets.</li><li>Reduced training resource usage up to 10x while minimally impacting test performances.</li><li>Analyzed pathogenicity prediction on variants of uncertain significance (90% consistency across 72 models).</li><li>Preparing research paper for IEEE EMBC 2026 submission (Jan 2026).</li></ul> <div><div>ML</div><div>Genomics</div><div>Python</div><div>Hydra</div><div>Scikit-learn</div></div> |
| Sep 2025<br>Mar 2025 | <b>Student Researcher   FPGA Deep Space, NECSTLAB, POLITECNICO DI MILANO, Italy</b> <ul style="list-style-type: none"><li>Accelerated ECOS library for autonomous navigation of deep space satellites.</li><li>Developed accelerator for sparse matrix computations (SpMV and LDLt kernels) using Vitis HLS and Vivado.</li><li>Deployed IP on Kria KD240 and developed a C Linux driver, achieving 2x speedup over software.</li><li>Competed in the 2025 edition of the AMD Open Hardware competition.</li></ul> <div><div>FPGA</div><div>HLS</div><div>Vitis</div><div>Vivado</div><div>Xilinx</div><div>Embedded systems</div><div>C</div></div>                                                                          |
| May 2024<br>Mar 2023 | <b>Software Engineer Intern   Formula SAE, MORE MODENA RACING, UNIMORE, Italy</b> <ul style="list-style-type: none"><li>Co-led a team of 4 developing software for combustion and hybrid Formula SAE cars.</li><li>Developed live, wireless telemetry (1 Mbps over 500m) using Raspberry Pi 5 and Grafana.</li><li>Developed car dashboard software on STM32 controller reading CAN bus data.</li></ul> <div><div>Automotive</div><div>C</div><div>Raspberry Pi</div><div>Live telemetry</div><div>Grafana</div><div>STM32</div><div>CAN</div></div>                                                                                                                                                                          |

PROJECTS

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| <b>STM32L4 IoT DRIVERS</b><br> <a href="https://github.com/SamueleTondelli/stm32l4-discovery-kit-iotnode">github.com/SamueleTondelli/stm32l4-discovery-kit-iotnode</a><br>Bare metal drivers for STM32L4 Discovery kit IoT, including I2C sensors and SPI WiFi module. <div><div>C</div><div>STM32</div><div>I2C</div><div>SPI</div><div>WiFi</div></div> | 2023 |
| <b>STEAM SEARCH ENGINE</b><br> <a href="https://github.com/RaffaeleTranfaglia/Steam-Search-Engine">github.com/RaffaeleTranfaglia/Steam-Search-Engine</a><br>Search engine for 26k+ Steam games using Whoosh with Transformer-based sentiment analysis. <div><div>Python</div><div>Whoosh</div><div>Transformer</div><div>NLP</div></div>                  | 2023 |

LANGUAGES

|         |       |
|---------|-------|
| Italian | ●●●●● |
| English | ●●●●○ |

+ STRENGTHS

- Curiosity
- Motivation
- Passion